

Task: End-to-End Data Management Pipeline for RecoMart Recommendation System

Course: Data Management for Machine Learning

Assignment: I

Date: 1/21/2026

Problem Formulation

1. Business Problem Definition

Primary Objective:

The primary objective for **RecoMart** is to design and implement a data-driven recommendation engine to enhance **customer engagement** and **cross-selling opportunities**². By personalizing the user experience, the system aims to improve sales conversion rates and retain customers on the platform³³³³. The pipeline must be automated and modular to ensure that the recommendation model consistently learns from **fresh, validated, and well-structured data**⁴.

Business Challenge:

RecoMart, an e-commerce startup, faces the following challenges:

- Customers experience choice overload with thousands of products
- Low engagement rates due to generic product displays
- Missed cross-selling and upselling opportunities
- Suboptimal conversion rates from browsing to purchase

Solution Approach:

Build a data-driven recommendation engine that leverages user behaviour patterns, transaction history, and product metadata to deliver personalized product suggestions that match individual user preferences and increase purchase likelihood.

2. Key Data Sources and Attributes

The system will ingest data from multiple sources to capture a 360-degree view of user behavior and product characteristics⁵⁵⁵⁵:

Data Source	Type	Key Attributes
Clickstream Logs	Behavioral	Log ID, User ID, Product ID, Action (click/view), Timestamp, Device Type
Transactional History	Historical	Transaction ID, User ID, Product ID, Purchase Amount, Date , Quantity
Product Metadata	Catalog	Product ID, Product Name, Category, Price, Rating, Inventory Status, Brand
External APIs	Contextual	Product ID, Sentiment scores (reviews), Product popularity Index

3. Expected Pipeline Outputs

To support the end-to-end ML workflow, the pipeline will deliver the following outputs

3.1 Clean Datasets for Analysis

Curated data free of missing values and duplicates, suitable for analysing interaction distributions and sparsity patterns

3.2 Engineered Features

Specialized features such as user activity frequency, average ratings per item, and similarity-based features for collaborative or content-based models

3.3 Trained Recommendation Models

- **Collaborative Filtering Model:** Matrix Factorization (SVD/ALS) for personalized recommendations
- **Content-Based Model:** Item similarity based on product attributes
- Model artifacts with versioning and metadata tracking
- Performance benchmarks and comparison reports

3.4 Inference Interface

- API endpoint for real-time recommendation requests
- Batch prediction pipeline for offline recommendations
- Feature retrieval mechanism from feature store
- Logging and monitoring for recommendation quality

4. Evaluation Metrics

The recommendation system will be evaluated using the following metrics:

Metric	Description
Precision@K	Measures the proportion of recommended items in the top-K set that are relevant to the user
Recall@K	Measures the proportion of relevant items that are successfully captured in the top-K recommendations
NDCG@K	NDCG (Normalized Discounted Cumulative Gain): Evaluates the ranking quality by accounting for the position of relevant items in the recommendation list
Coverage	Percentage of items that can be recommended
Diversity	Variety of categories in recommendations

Business Success Metrics:

- Click-through rate (CTR) on recommendations
- Conversion rate improvement
- Average order value (AOV) increase

Conclusion

This problem formulation establishes the foundation for building a comprehensive data management pipeline for RecoMart's recommendation system. The pipeline will process multi-source data through automated ingestion, validation, transformation, and model training workflows, ultimately delivering personalized product recommendations that drive business value through improved customer engagement and conversion rates.